



Samueli
Bioengineering



Development of Electromyography-Controlled Assistive Robotics for Self-Feeding

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Advisor: Dr. Veronica Santos



Agenda



Introduction/Background



Gaps in Knowledge



Methods and Experimental Design



Results and Discussion



Conclusion and Future Directions

Disability in the United States

- About 1 in 8 Americans have a mobility disability
- Dependency on caregivers can lead to feelings of shame and lower self-worth
- Assistive robots aims to improve independence in daily tasks



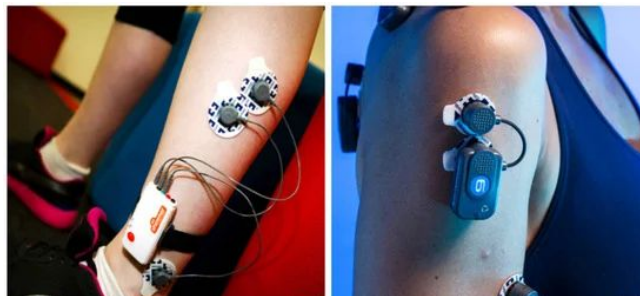
(CDC)

Accessible Controls for Assistive Robotics

- Current input systems have limitations
 - Keyboard & mouse, joystick
 - Often require precise movement
- **Electromyography as an alternative**
 - Detect muscular contractions without precise movement
 - Any activatable muscle



<https://graspat.com/shop/access/joysticks/bjoy-ring/>



Al-Ayyad et al, *Electronics*, 2023

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Gaps in Existing Assistive Robotics

KINOVA 'JACO' Arm



<https://top3dshop.com/product/kinova-jaco-assistive-robotic-arm>

- Many tasks
- Confined space
- ✓ Assistive eating

'Obi' Tabletop Robot



<https://mortonterry.co.nz/product/obi/>

- One task
- Confined space
- ✓ Assistive eating

'Stretch' Mobile Manipulator



V Nguyen

- Many tasks
- Independently mobile
- ✗ Assistive eating

Objective for Project and Significance

- Develop a user-friendly and open-source assistive feeding system for Stretch 3
- Incorporate user-specific EMG sensor control for users with limited mobility



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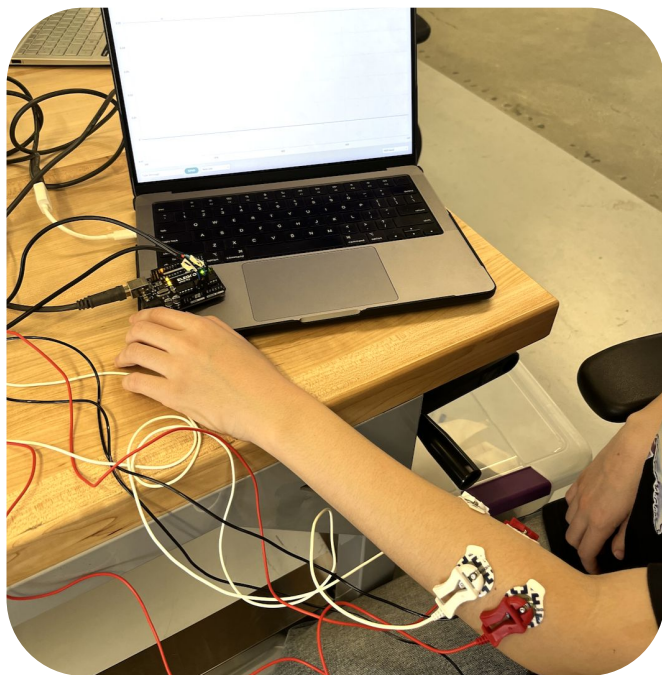


Results and Discussion

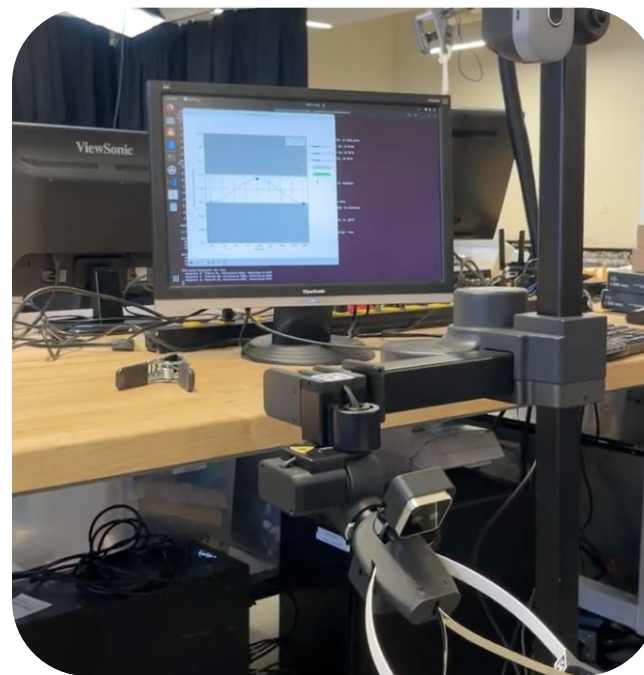


Conclusion and Future Directions

Two Development Subsystems

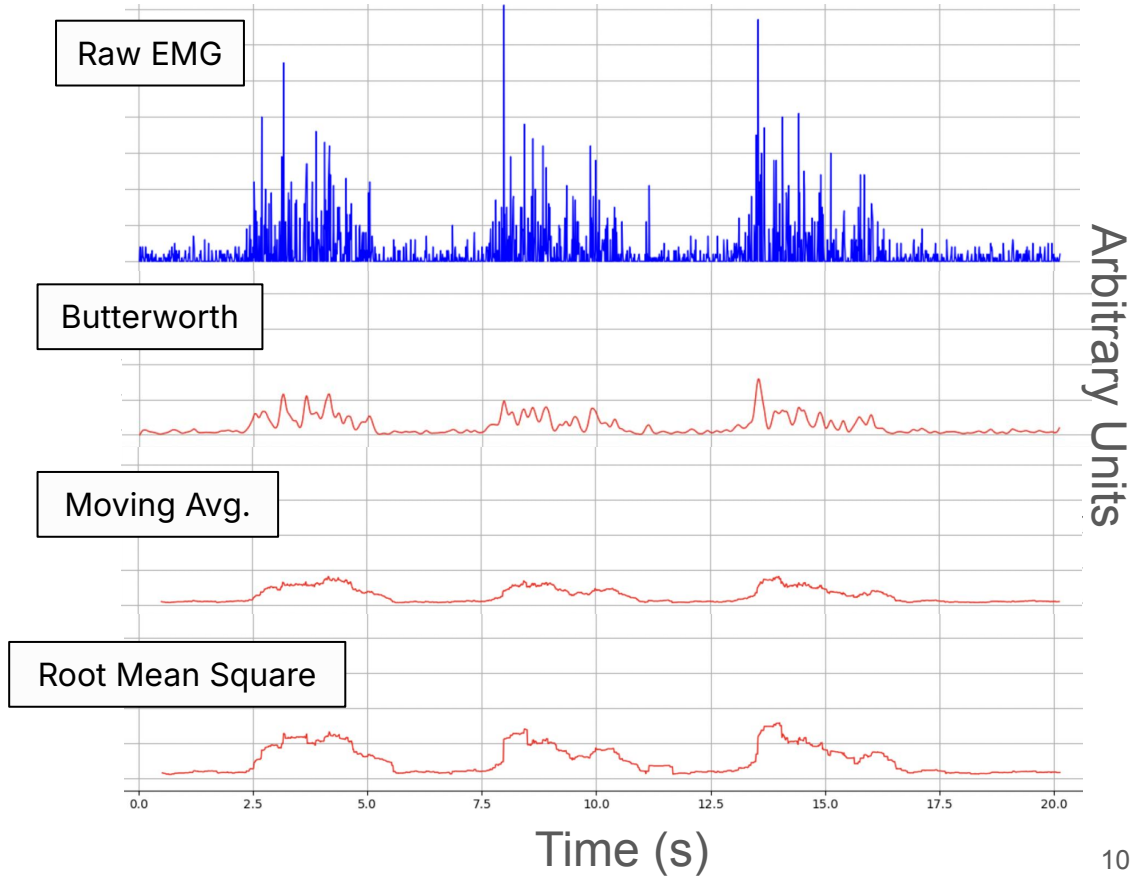
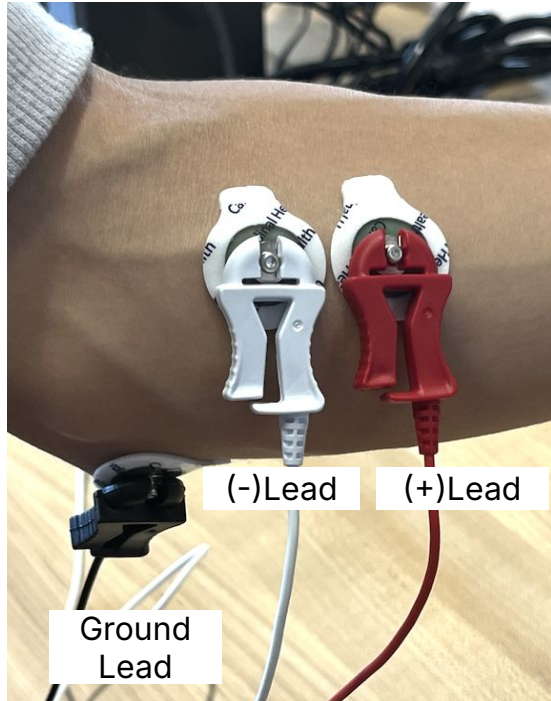


Electromyography

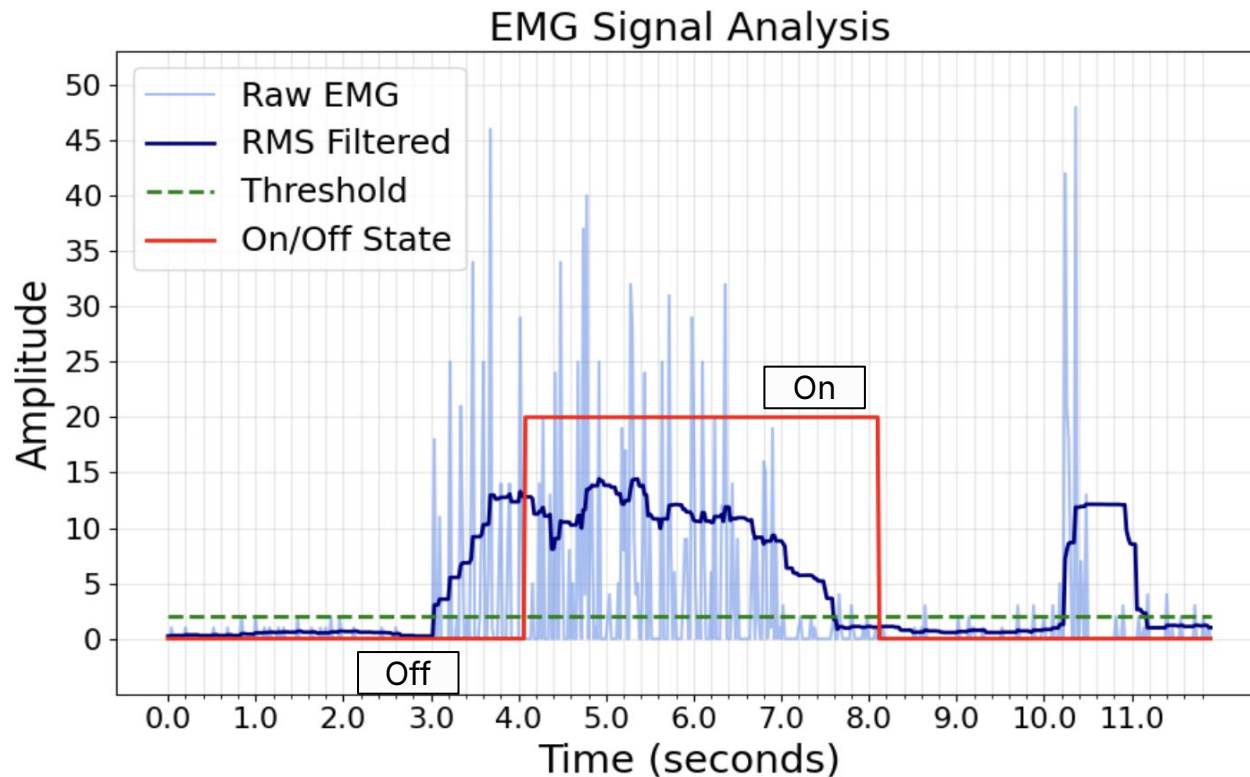


Robot Body

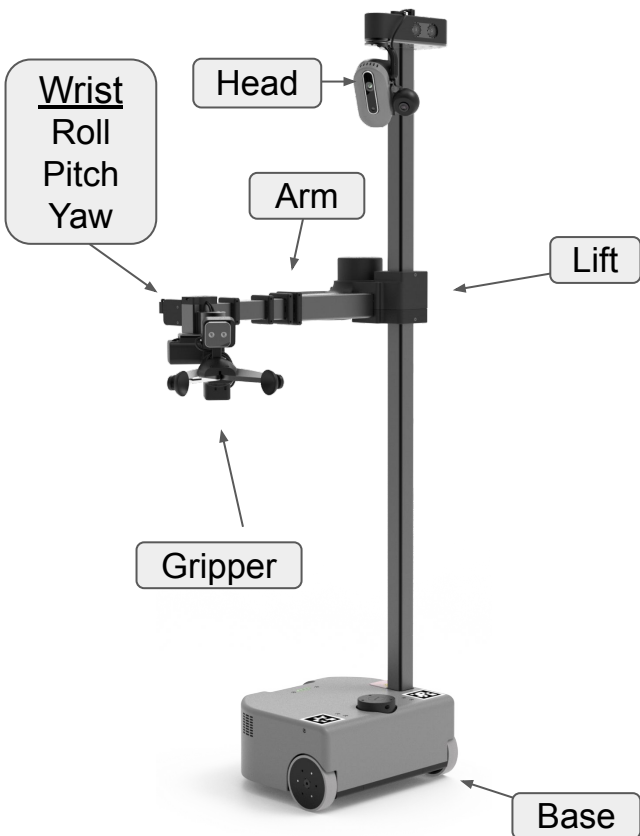
EMG Setup and Digital Filter Selection



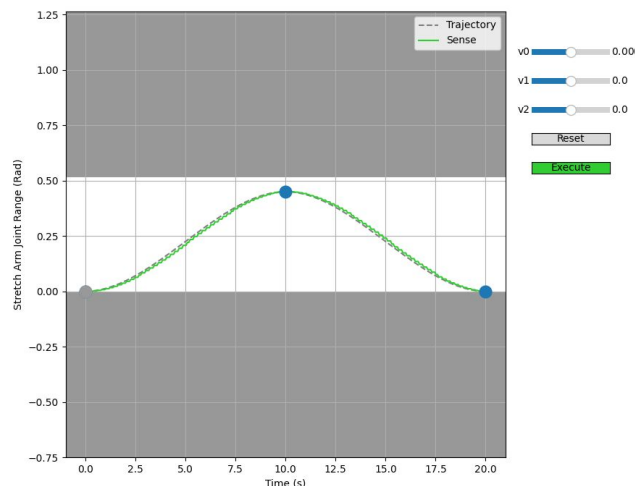
EMG Signal Processing for the Stretch Robot



Trajectories Program Individual Joint Movements



Cubic Spline Trajectory

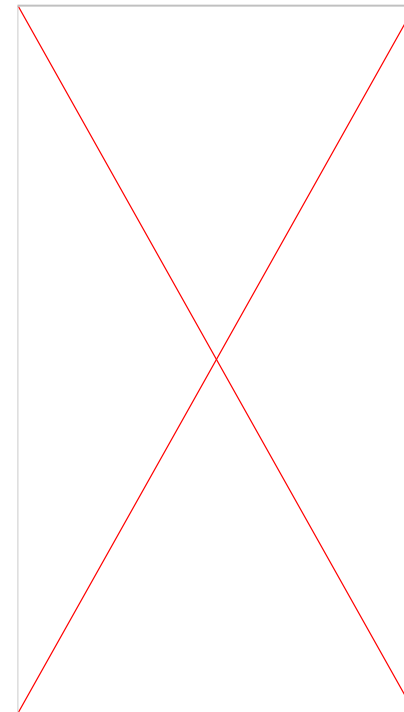
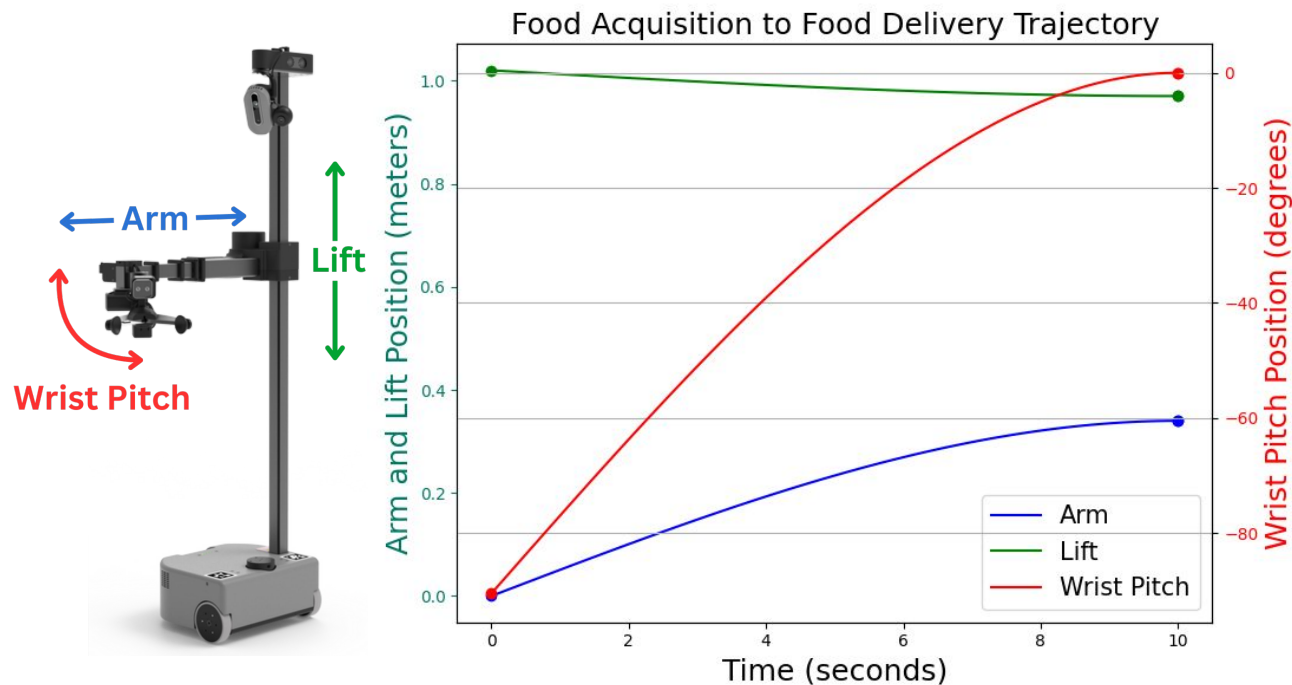


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Extracted Waypoints for arm:
Waypoint 0: Time=0.00, Position=-0.0000, Velocity=-0.0003
Waypoint 1: Time=10.00, Position=0.4500, Velocity=0.0000
Waypoint 2: Time=20.00, Position=0.0000, Velocity=0.0000
```

Three Full Body Trajectories

- Food Acquisition
- Food Delivery
- Returning to Plate

Trajectory Planning Using Waypoints



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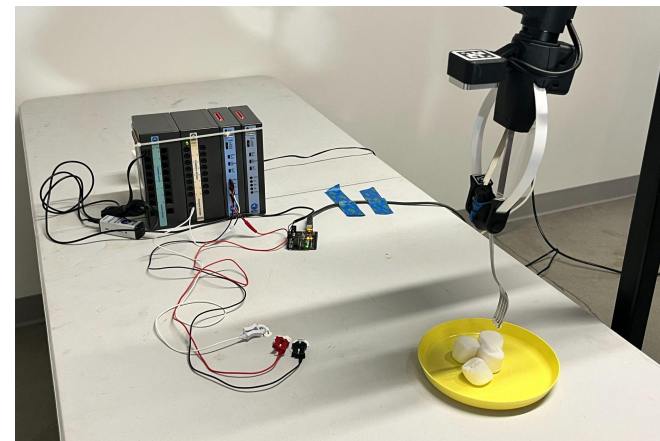
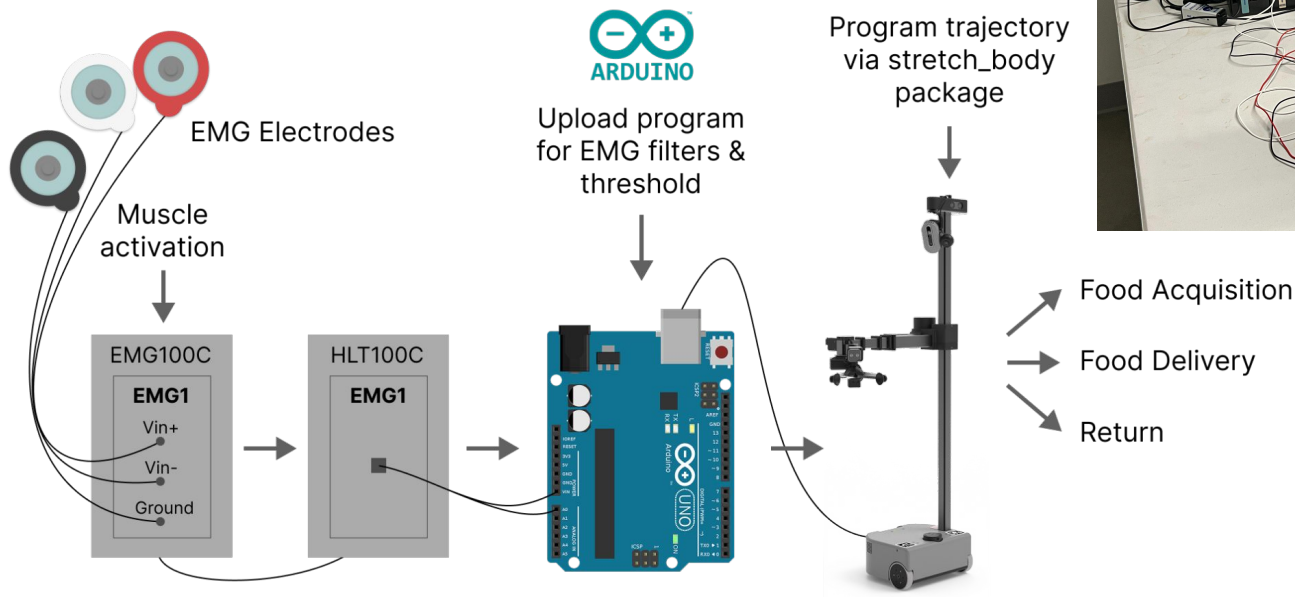
Results and Discussion



Conclusion and Future Directions

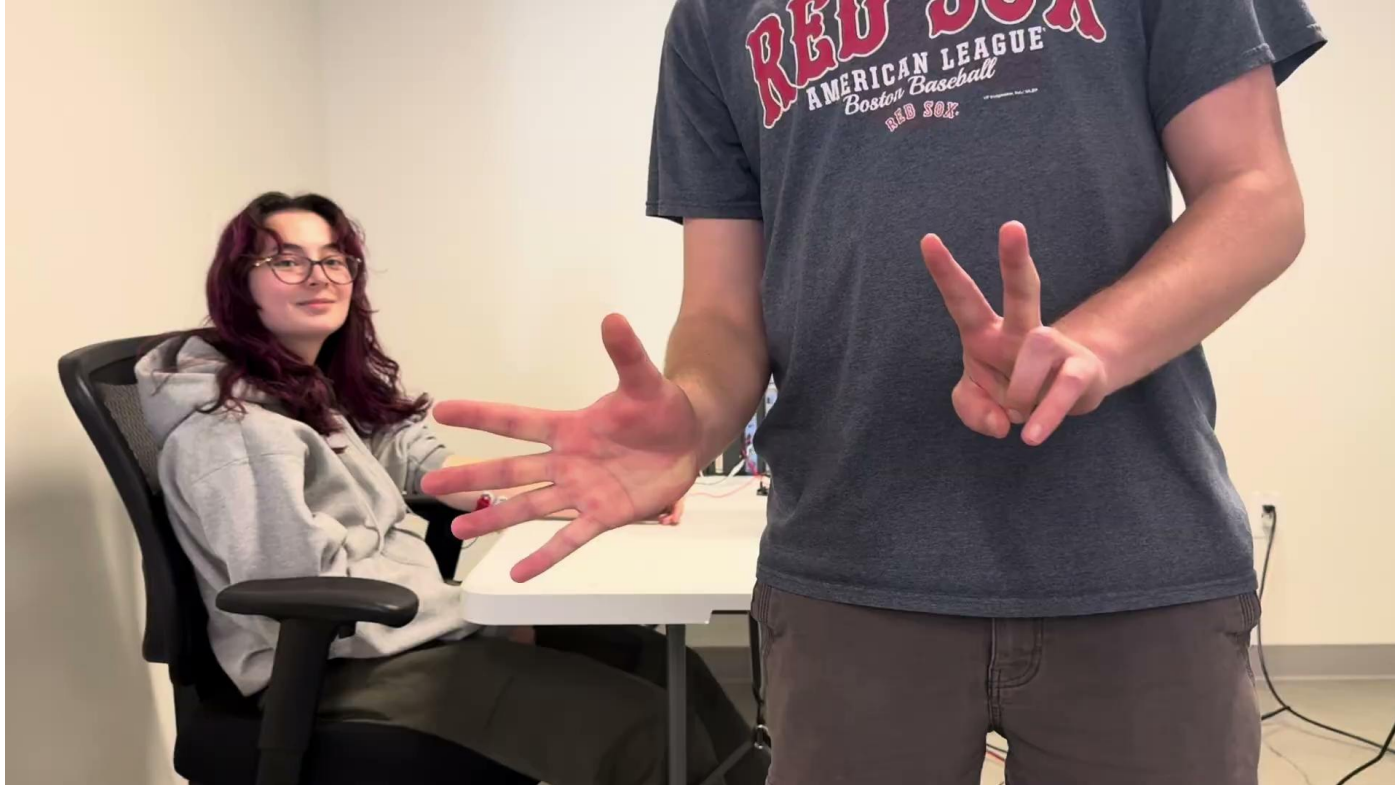
Final System Overview for Assistive Feeding

A. System schematic



B. Final Setup

Final Assistive Feeding System Demonstration



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Significance of Advancing EMG-Controlled Assistive Feeding

- EMG addresses limitations of other input methods
- Adaptable to different muscular activation capabilities
- Expand functionality for Stretch 3
- Providing open-source program can impact many people
- Potential for functional independence while eating



Expanding Capabilities & Usability in Our Assistive Technology



Hardware:

Add second EMG input for more precise control.
Improve end effector for better food acquisition.
Implement an emergency stop for safety.



Software:

Enhance EMG processing for smoother control.
Develop a customizable, user-friendly GUI.
Improve response time for real-time adaptation.



User Testing:

Conduct trials with diverse participants.
Refine usability based on real-world feedback.

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References

- Ackerman, E. (2023). This Robot Could Be the Key to Empowering People With Disabilities. *IEEE Spectrum*.
- Bell, B. (2022). Home Is Where the Robot Is. *Illinois Advancement Storied Magazine*, University of Illinois Urbana-Champaign.
- Brose, M., Halpern, M., & Reed, K. (2022). Assistive technology and independence: A review of mental health outcomes for individuals using feeding aids. *Journal of Rehabilitation and Assistive Technology*, 28(3), 234-245.
- Buder, S. & Perry, H. The Social Model of Disability Explained. Apr 12, 2024
- Campeau-Lecours, A., et al. (2017). Kinova Modular Robot Arms for Service Robotics Applications. *International Journal of Robotics Applications and Technologies*, 5, 49-71.
- Hermann, R. P., et al. (1999). Powered feeding devices: An evaluation of three models. *Archives of Physical Medicine and Rehabilitation*, 80(10), 1237-1242.
- Jones, A., & Miller, D. (2023). Technology in elder care: Bridging the caregiver gap through assistive devices. *Journal of Geriatric Care Innovation*, 12(2), 89-102.
- Kim, S., & Wang, H. (2021). The impact of assistive devices on social engagement and mental well-being in individuals with disabilities. *Disability and Society*, 36(7), 945-961.
- Lee, T., Zhang, R., & Singh, P. (2020). Challenges in adaptive control and intent recognition for assistive robotics. *International Journal of Robotics Research*, 39(5), 456-478.
- Losey, Dylan P., et al. "Controlling assistive robots with learned latent actions." 2020 IEEE International Conference on Robotics and Automation (ICRA), May 2020
- Naotunna, I., et al. (2015). Meal assistance robots: A review on current status, challenges, and future directions. *2015 IEEE/SICE International Symposium on System Integration (SII)*, 211-216.
- Olatunji, S. A., Nguyen, V., Cakmak, M., Edsinger, A., Kemp, C. C., Rogers, W. A., & Mahajan, H. P. (2024). Immersive participatory design of assistive robots to support older adults. *Ergonomics*, 67(6), 717-731.
- Smith, J., Lopez, A., & Johnson, E. (2021). The aging caregiver workforce and implications for long-term care support. *Health Affairs*, 40(10), 1803-1810.

Thank You!

